# Douglas Miller

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I am a fourth-year undergraduate with research experience in CMP theory and quantum gravity. I intend to pursue a PhD in CMP theory. I plane to focus my PhD research primarily on topological phases and matter, anyonic physics, and quantum Hall interferometry. My long-term goal is to lead a research group as a theoretical physicist, with a commitment to understanding novel phenomena using creative mathematics while training the next generation of researchers to be even better than ours.

## Education

**Florida Atlantic University (FAU)** Boca Raton, Florida
*B.S. in Physics, minor in Mathematics* Expected May 2027

- Cumulative GPA: 3.98/4.00
- Senior honors thesis (in progress): *Quantum Gravity: Continuum Reconstruction from Discrete Ponzano–Regge Data*.

## Research Experience

**Florida Atlantic University, Department of Physics** Boca Raton, Florida
*Undergraduate Researcher* Fall 2025–Present

Advisor: Prof. Jonathan Engle | jengle7@fau.edu

- Project: Levi-Civita Curvature in Ponzano-Regge.
- Investigating tension between two different representations of the Ponzano-Regge spinfoam model of 3D Euclidean quantum gravity. Developed code to compute spinfoam amplitudes and expectation values of geometrical observables, such as Regge deficit-angle. Manuscript in progress.

**University of Rochester, Department of Physics and Astronomy** Rochester, New York
*Undergraduate Research Intern | NSF REU and current collaboration* Summer 2026–Present

Advisor: Prof. Chaitanya Murthy | cmurthy@ur.rochester.edu

- Project: scattering theory for quantum point contacts in quantum Hall Fabry–Pérot interferometry.
- Developed a gauge-independent treatment of electron scattering theory for quantum Hall interferometry, extendable to more general scattering potentials in quantum Hall interferometers.

## Talks and Presentations

**Talk and Poster presentation | University of Rochester Physics REU Symposium** Summer 2026
*Gauge-Invariant Treatment of Electron Scattering for Quantum Hall Interferometry*

**Contributed talk | 12th Gulf Coast Gravity Meeting, Troy, Alabama** April 17–18, 2026
*Expectation Value of Levi-Civita Curvature in Ponzano-Regge: Non-flatness in 3D Quantum Gravity*

**Research Collquium | Department of Physics, FAU** April 2026
*Expectation Value of Levi-Civita Curvature in Ponzano-Regge*

**Dr. Eric Shaw Excellence in Science Undergraduate Research Poster Competition | FAU** October 2025
*Curvature in Ponzano–Regge 3D Quantum Gravity*

## Honors and Awards

**Bjorn Lamborn Scholarship in Physics** Spring 2026
- Excellence in Theoretical Physics

**Florida Atlantic University Honors Program** 2023–2024
- Completed the two-year program in one year.

**President's List** Fall 2023, Spring 2025, Fall 2025, Spring 2026
- Awarded for attaining a semester GPA of 4.00.

**Dean's List** Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026
- Awarded for attaining a semester GPA of at least 3.75.

## Teaching and Outreach

**Physics and Mathematics Tutor** August 2024–Present
- Tutor high-school and college students in calculus and introductory physics courses. Pre-calculus, Calculus 1, Physics 1,2.

**Annual Pumpkin Drop and Physics Carnival, FAU Department of Physics** October 2023–Present
- Assist with event preparation and outreach activities. Help host 400 4th, 5th, and 6th graders.

**Loops '24 International School and Conference on Loop Quantum Gravity** Spring 2024
- Provided volunteer and logistics support for the international school and conference.

## Technical Skills and Languages

**Computer programming:** Python (NumPy, SciPy, Matplotlib, SymPy), Mathematica, HTML, CSS

**Typesetting:** LaTeX

**Languages:** English (fluent), Polish (fluent), Spanish (elementary), German (elementary)

## Relevant Course-Work

**Graduate Level:** Mathematical Physics 1 (functional analysis) | Quantum Mechanics | Algebraic Topology | Classical Mechanics

**Undergraduate Level:** General Relativity (directed independent study) | Complex Analysis | Discrete Mathematics